

JobQuest

Report — Chapters 1 to 5

Introduction, Background, Related Work, Proposed System, and Analysis

An AI-native experiential career-exploration platform

Specialized Project (ĐACN)

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Abstract

Occupational choice is commonly made with limited direct knowledge of the work involved. Research on career decision-making identifies insufficient occupational information as a principal source of difficulty, and the realistic job preview literature establishes that balanced, candid previews improve person-job fit and reduce subsequent regret. Delivering such previews has historically required either workplace exposure or content authored occupation by occupation, neither of which scales. This project designs, implements, and evaluates a platform that uses generative language models to deliver experiential occupational exposure at greater breadth, and establishes whether competence demonstrated within such an experience can be measured with sufficient reliability to be useful.

The report establishes the evidence base for this approach, reviews existing research prototypes and deployed systems against criteria derived from properties the literature identifies as consequential for learning systems, and specifies a proposed platform in which occupations are presented as performable tasks, responses are assessed against expert-authored rubrics, and demonstrated competence determines subsequent exploration. The review finds that the constituent techniques have been demonstrated separately but not in combination, and that existing systems in this area report assessment agreement without chance correction or difficulty stratification. The proposed evaluation addresses this through an expert-reviewed reference set constructed within a single occupational field. The report covers Chapters 1 to 5.1: introduction, background, related work, the proposed system, and its analysis.

Keywords: career exploration, experiential learning, realistic job preview, large language models, automated assessment, stealth assessment, serious games.

List of Abbreviations

Abbreviation	Expansion
BR	Business Rule
DR	Data Requirement
ECD	Evidence-Centered Design
ESCO	European Skills, Competences, Qualifications and Occupations
FR	Functional Requirement
HCMUT	Ho Chi Minh City University of Technology
LLM	Large Language Model
MBTI	Myers-Briggs Type Indicator
NFR	Non-functional Requirement
ONET	Occupational Information Network
RIASEC	Realistic, Investigative, Artistic, Social, Enterprising, Conventional
RJP	Realistic Job Preview
RP	Research Problem
SCCT	Social Cognitive Career Theory
SDT	Self-Determination Theory
UC	Use Case
ĐACN	Đồ án chuyên ngành (Specialized Project)

1 Introduction

1.1 Motivations

Occupational choice is among the most consequential decisions an individual makes, and it is typically made with limited direct knowledge of the work itself. A prospective entrant selects a field on the basis of job titles, salary figures, and descriptions authored by parties with an interest in recruiting them. The texture of the work — its daily pressures, its trade-offs, the tasks that occupy most of its hours — generally becomes apparent only after entry, when the cost of reversing the decision is high.

Research on career decision-making identifies the mechanism behind this difficulty. In the validated taxonomy of Gati, Krausz and Osipow [1], subsequently cross-validated across thirteen countries, insufficient information about occupations constitutes a principal category of decision difficulty, distinct from indecisiveness or inadequate self-knowledge. The obstacle is informational rather than dispositional: individuals are not primarily unable to decide, but unable to decide on an adequate basis.

Conditions in Vietnam illustrate the problem in a particularly compressed form. Entry to higher education is mediated by national examination results and published admission benchmark scores, so the selection of a field of study is effectively completed at seventeen or eighteen, under time pressure, and is constrained by the applicant's score relative to institutional thresholds rather than by any assessment of occupational fit [2]. The decision is frequently not the applicant's own. A 2026 survey of approximately 9,200 students across ten Vietnamese universities reported that more than one fifth had selected their field on parental direction, that nearly one fifth had entered without a defined occupational goal, and that around a third were subsequently uncertain whether they would choose the same field again [3]. Labour-market data indicate the downstream cost. Analysing successive waves of the national Labour Force Survey, Tran et al. [4] found the proportion of graduates in fully matched employment declining over time, with vertical and combined mismatch carrying measurable wage penalties; related analyses associate the same pattern with underutilised human capital at national scale. Employer-reported skills shortages, particularly in science and technology fields, compound the effect.

The prevailing response, in Vietnam and elsewhere, is psychometric. Interest inventories derived from Holland's typology, personality instruments in the Myers-Briggs tradition, and their commercial derivatives dominate career guidance provision. These instruments are informative about preferences but structurally unable to address the difficulty Gati's taxonomy identifies: they elicit what an individual believes about themselves rather than supplying information about occupations, and they cannot establish whether that individual would perform or persist in a given role. Their measurement properties also invite caution — an estimated 39 to 76 per cent of respondents are assigned a different Myers-Briggs type on retest after five weeks [5].

An alternative is well established in organisational psychology but rarely applied to career exploration. The realistic job preview literature demonstrates that balanced previews, including a role's unattractive features, improve person-job fit and reduce subsequent turnover, with perceived honesty identified as the opera-

tive mechanism [6][7]. Such previews are, in effect, information about the work delivered before commitment. Their delivery, however, has historically depended on either direct workplace exposure or content authored case by case, neither of which scales to the breadth of occupations an undecided student might wish to consider. Recent advances in generative language models alter this constraint, and that change in feasibility motivates the present project.

1.2 Objectives

The project designs, implements, and evaluates a platform that delivers experiential career exploration at a scale authored content cannot reach, and in doing so establishes whether competence demonstrated within such an experience can be measured reliably enough to be useful. Its objectives are:

1. To determine how AI-generated occupational scenarios can be constrained so that they remain faithful to the occupations they represent, and to specify an authoring method that achieves this.
2. To establish whether an automated assessor can evaluate open-ended responses to such scenarios with sufficient agreement with domain experts to support formative use.
3. To design a progression model in which competence demonstrated in one occupation informs which adjacent occupations a user is prepared to explore.
4. To determine which game-design mechanisms sustain engagement in career exploration without displacing the learning they are intended to support.
5. To construct an evaluation methodology, based on an expert-reviewed reference set, capable of substantiating or refuting the preceding claims offline.

These objectives are realised in a deployable platform, JobQuest, specified in Chapter 4. The present phase delivers a working system covering the information technology field; the subsequent capstone phase extends its occupational coverage and operational maturity. The present chapters establish the evidence base for its design and its position relative to existing work.

1.3 Scope

The system delivered in this phase is complete in function but bounded in coverage to the information technology field and its specialisations. The restriction is methodological rather than incidental: constructing a reference set of gold-standard scenarios and expert judgments requires domain expertise that the project possesses only in this field, and without such a set the second and fifth objectives cannot be tested. Generalisation to other occupational domains is treated as subsequent work, and the design is intended to be domain-agnostic even though the evaluated instance is not.

Within this boundary the work covers scenario generation and adaptation, automated assessment of responses, a competence-based progression model, mechanisms for sustaining engagement, an authoring capability for domain experts, and the offline evaluation. It does not extend to longitudinal study of career outcomes, which would require a time horizon beyond the project, nor to recruitment functions: the system does not broker applications or transmit user performance to employers. Personality and interest assessment is retained only in a supporting capacity. The project does not claim that simulated experience substitutes for employment; the available evidence does not support such a claim, and the system is

positioned accordingly.

1.4 Significance of the Project

1.4.1 Significance from the Practical Perspectives

The practical contribution is a means of acquiring occupational information experientially before commitment, applicable to students selecting a field of study, to workers considering transition, and to career guidance provision more broadly. Its value derives from three established findings: that balanced, honest previews improve fit and reduce later regret [6][7]; that the obstacle they address — insufficient occupational information — is a principal source of career decision difficulty [1]; and that successful performance of authentic tasks raises task-specific self-efficacy, which social cognitive career theory identifies as a determinant of choice and persistence [8][9]. For the Vietnamese context in particular, it offers an alternative to a guidance market composed almost entirely of self-report instruments, at a decision point where applicants have little opportunity to observe the work they are selecting.

1.4.2 Significance from the Scientific Perspectives

The scientific contribution lies in combining and validating techniques that have been demonstrated separately but not together: constrained generation of domain-accurate scenarios, unobtrusive assessment of competence from behaviour, and automated judgment of open-ended responses. Each carries a documented weakness — drift and factual instability in open-ended generation, and position, verbosity and self-preference bias in automated judging — and the project’s methodological contribution is an evaluation design that measures these directly rather than assuming them resolved. The existing prototypes in this area are predominantly evaluated through small qualitative or within-subject studies; an evaluation based on an expert-reviewed reference set, reporting chance-corrected agreement and difficulty-stratified performance, offers a more reproducible basis for the claims such systems make.

1.5 Report Structure

Chapter 2 establishes the background: the evidence that experiential exposure conveys occupational information more effectively than description, the basis for measuring competence from behaviour, the conditions under which game-based delivery supports rather than displaces learning, relevant career-development theory, the generative and assessment technologies involved, and the data resources and benchmarks available for grounding and evaluation. Chapter 3 reviews related work, examining research prototypes and deployed systems, deriving criteria for comparison, and identifying the gap the project addresses. Chapters 4 to 8 follow the prescribed structure for the specialized project, covering the proposed system, its analysis and design, implementation and testing, evaluation, and conclusions.

2 Background Knowledge and Technologies

This chapter assembles the evidence on which the proposed design rests. Sections 2.1 and 2.2 concern learning and measurement: whether performing a task conveys understanding that description cannot, and whether competence can be inferred from performance. Section 2.3 examines game-based delivery, including evidence qualifying its effectiveness. Section 2.4 summarises career-development theory. Sections 2.5 and 2.6 address the generative and assessment technologies and the data resources available for grounding and evaluation, and Section 2.7 considers responsibility constraints arising from the intended user population.

2.1 Experiential learning and realistic job previews

Kolb's model of experiential learning [10] characterises learning as a cycle in which concrete experience is transformed through reflective observation into abstract conceptualisation and subsequent active experimentation. The model's implication for occupational information is direct: an occupational description supplies conceptual content without the experience from which, on this account, durable understanding is constructed. Whatever a description conveys, it does not engage the transformation process the model describes.

The realistic job preview literature provides the corresponding empirical tradition [11]. A realistic job preview presents a balanced account of a role, including features an applicant might find unattractive, in place of a promotional account. Four meta-analyses converge on its effects [6][7][12][13]. Phillips [6], synthesising forty studies, associated such previews with reduced inflation of expectations, increased self-selection, improved performance and lower voluntary turnover, and additionally found richer audiovisual previews more effective than written ones — a finding that bears directly on delivery format. Earnest, Allen and Landis [7], in a path analysis over fifty-two studies and approximately seventeen thousand participants, identified perceived honesty as the mediating mechanism, indicating that the effect depends on the preview's candour rather than merely on its informational content.

Whether simulated experience transfers to real performance is addressed by the simulation-based learning literature. Cook et al. [14] and McGaghie et al. [15] established substantial effects for technology-enhanced simulation within health professions education, where the paradigm originated. Of greater relevance here, Chernikova et al. [16], synthesising 145 studies in higher education, reported a large overall effect that generalised across academic domains rather than remaining confined to clinical training, and further found that scaffolding — worked examples for novices, reflective prompts for more advanced learners — moderated outcomes.

Taken together, these three literatures support a specific proposition: that an honest, performed representation of a role conveys occupational information more effectively than description, and that the effect is not restricted to the clinical domains in which simulation research began.

2.2 Assessment of competence from behaviour, and competency-based progression

If a learner performs tasks rather than answering questions about themselves, the assessment problem changes accordingly. Evidence-centered design, developed for assessment within game environments under the term *stealth assessment* [17], addresses this by decomposing the problem into three linked models: a competency model specifying what is claimed to be measured, an evidence model specifying which observable behaviours constitute evidence for those claims and with what weight, and a task model specifying situations that elicit the relevant behaviour.

Table 2.1: Evidence-centered design models and the requirements they impose.

Model	Function	Requirement it imposes
Competency	Defines the constructs claimed to be measured	Constructs must be specified independently of any single task if inferences are to transfer
Evidence	Maps observed behaviour to inferences about the constructs	Each observable must be attributable to a named construct; unattributable observations yield uninterpretable scores
Task	Specifies situations that elicit relevant behaviour	The target construct must be necessary to task completion, not merely permitted by it

The third requirement is the most demanding and the most frequently unmet. Shute et al. [18] obtained significant correlations between in-game estimates of problem-solving and external measures precisely because the tasks could not be completed without deploying the measured competency. Where a task admits a route that bypasses the target construct, the evidence model measures something other than what it claims. Task authoring in such systems is consequently a psychometric activity rather than a purely editorial one.

A related question is what an assessment result entitles a learner to do next. Competency-based progression [19] treats competence as a transferable and independently verifiable unit rather than as an attribute of a completed course or an occupied role. On this account, competence demonstrated in one context constitutes evidence bearing on readiness for adjacent contexts. Establishing which contexts are genuinely adjacent requires an external basis, and standard occupational taxonomies supply one by describing occupations in terms of constituent skills and work activities, discussed in Section 2.6.

2.3 Game-based delivery: evidence, moderators, and limits

Game-based delivery is frequently proposed for educational systems and frequently overstated. The evidence supports it conditionally, and the conditions are specific enough to constrain design.

Effects and moderators. Hamari, Koivisto and Sarsa [20] found gamification effects generally positive but dependent on context and user. Sailer and Homner [21], in the principal learning-specific synthesis, reported significant but modest effects — cognitive $g = .49$, motivational $g = .36$, behavioural $g = .25$ — and identified narrative and the combination of competition with collaboration as moderators, noting the effectiveness of narrative paired with a persistent developing avatar. Bai, Hew and Huang [22] and Clark, Tanner-Smith and Killingsworth [23] report comparable medium effects with design features moderating outcomes.

The competence qualification. A meta-analysis by Li, Hew and Du [24] disaggregates these effects by psychological need and yields a result that materially constrains claims made for such systems: gamification produced large gains in perceived autonomy ($g = 0.638$) and relatedness ($g = 1.776$) but only a marginal effect on competence ($g = 0.277$). Gamification, in other words, reliably improves how learners feel about an activity while doing comparatively little for what they can do. Zeng et al. [25] add that particular element combinations underperform, with one common configuration associating negatively with academic performance. Any system asserting skill development must therefore locate the mechanism of that development in something other than reward structures.

Evidence by genre. Genre-level findings differentiate more usefully than aggregate ones. Role-play methods show an overall effect of approximately 0.82 with the largest component on skill acquisition [26], the strongest genre-level support for role-based designs. Simulation games carry the most occupationally relevant evidence and the clearest caveat: Sitzmann [27], synthesising sixty-five studies, found simulation-taught trainees exceeded comparison groups on self-efficacy, declarative and procedural knowledge and retention, but only where the game actively conveyed material, permitted learner control, and supplemented other instruction — when it constituted the sole method, comparison groups performed better, and the analysis reported evidence of publication bias. Wouters et al. [28] similarly found serious games effective for learning and retention while not necessarily more motivating than conventional instruction, and most effective as a supplement across multiple sessions. Narrative and branching-story formats reinforce engagement and learning [29], consistent with the narrative moderator above. Procedurally varied and randomised formats support replayability, but the evidence here is double-edged: randomised-reward mechanisms are robustly associated with problem gambling and reduced wellbeing among purchasers [30], which distinguishes variation used for content variety from variable-ratio reward scheduling.

Motivational basis. These findings are commonly interpreted through self-determination theory [31], under which motivation depends on satisfaction of needs for competence, autonomy and relatedness. The mapping is reasonably direct: progression and mastery structures address competence, meaningful choice addresses autonomy, and social or comparative features address relatedness, the least studied of the three in game research.

The implication for design is that engagement mechanisms and learning mechanisms should not be conflated. Narrative, progression and choice are supported as engagement structures; skill acquisition, on this evidence, must derive from the authenticity of the tasks themselves.

2.4 Career-development theory

Four theoretical traditions inform career guidance provision and the design of systems supporting it. Holland's typology [32] models occupational choice as person-environment congruence and underlies the interest inventories in widespread use, including the public-domain instrument distributed with the O*NET occupational database. Social cognitive career theory [8][9] identifies self-efficacy, built principally through mastery experience, as a determinant of interest formation and choice — a mechanism accessible to systems in which users perform tasks rather than report preferences, and inaccessible to instruments that only elicit self-description. Super's life-span theory [33] holds that vocational self-concept develops over time, so that occupational fit is properly understood as changing rather than fixed. Krumholtz's happenstance learning theory [34] extends this by treating unplanned events as a normal and potentially productive component of career development rather than as disruption.

The last two traditions jointly support a position that career guidance systems seldom represent: that a career decision is situated within a whole life, subject to circumstances — obligations, health, relationships, relocation, changing values — that alter what constitutes a suitable path. Where a system models occupational fit as a fixed verdict, this dimension is lost. The consideration carries particular weight in contexts where career decisions are made within strong family and social expectations, as the Vietnamese survey data indicate [3], and it provides the theoretical basis for treating work-life balance as an attribute of an occupation rather than as an externality.

The limitations of self-report instruments follow from the same body of theory. Beyond the retest instability noted in Section 1.1 [5], self-report is subject to social desirability effects, and congruence measures derived from interest typologies correlate only modestly with subsequent satisfaction. These are not implementation defects but properties of the method: asking what a person believes they would enjoy produces evidence of a different kind from observing what they do.

2.5 Generative and assessment technologies

Content generation. Automatic generation of educational content predates large language models but was long confined to shallow factual items [35]. Language models extended this substantially: Sarsa et al. [36] generated programming exercises with solutions, tests and explanations, a domain directly relevant to information technology scenarios, while noting that quality oversight remained necessary; Doughty et al. [37] generated items calibrated to specified cognitive levels. Controlled comparison indicates the outputs can be psychometrically adequate — one study found model-authored assessment items statistically indistinguishable from human-authored items on difficulty, discrimination and reliability, with the effect most pronounced in computer science [38]. The consistent qualification across this literature concerns grounding: fluent generation is readily obtained, while factual fidelity to a specified domain requires explicit constraint, typically through retrieval or authored scaffolding [39].

Interactive narrative and its control. Systems in which a language model conducts an interactive scenario draw on work demonstrating that such models can maintain memory, synthesise reflections and plan behaviour over extended interaction [40], and can track game state while a human retains narrative authority [41][42].

The same literature documents the principal failure mode, namely degradation of consistency over long horizons, and the mitigations that address it. Plan-driven generation frameworks report improved continuity and reduced spatiotemporal inconsistency relative to unconstrained prompting [43], indicating that authored structure rather than model scale is the operative variable.

Automated assessment of open responses. The use of language models as evaluators has developed through three identifiable stages, each imposing a different requirement on systems that adopt it. Zheng et al. [44] established feasibility, reporting agreement with human preferences above eighty per cent — comparable to inter-human agreement — while documenting position, verbosity and self-preference biases in the same work. Liu et al. [45] addressed method, showing that structured chain-of-thought prompting against an explicit rubric improves alignment, while noting a residual preference for model-generated text. The third stage concerns validation and is the most consequential. A systematic evaluation of twenty-one evaluators across approximately 541,000 judgments [46] found that raw agreement systematically overstates chance-corrected performance, reporting an approximately forty-one point discrepancy between exact-match agreement and Cohen’s kappa on one benchmark, and that high test-retest consistency can coexist with severe position bias. Dedicated benchmarks quantify the ceiling: on deliberately difficult response pairs with objective correctness labels, the strongest general-purpose evaluator reached approximately sixty-four per cent accuracy [47].

Four requirements follow for any system adopting this technique. Agreement must be reported with chance correction rather than as exact match. The evaluator should be drawn from a different model family than the generator, since self-preference otherwise contaminates the loop. Option ordering and response length must be controlled, since both bias scores independently of quality. And because accuracy degrades unevenly with item difficulty, aggregate figures conceal failure on precisely the reasoning-intensive items such systems most need to assess, which requires stratified reporting.

Expert involvement in authoring. Maintaining accuracy in generated instructional content is a multi-party problem. Recent work converges on a layered pipeline: a curated source of truth, retrieval-grounded generation, expert review prior to publication, and a feedback loop from use. Evaluation of such pipelines indicates that automated checks reliably enforce verifiable properties while domain experts remain necessary for pedagogical adequacy and for the distinctions that make an item discriminating [48]. The division of labour this implies — automated first-pass generation under expert validation — is the current best-supported arrangement for content that must remain factually accurate about a real domain.

2.6 Data resources and benchmarks

Systems making claims about occupations and about assessment accuracy can be grounded in, and evaluated against, established resources rather than bespoke data collection.

Occupational and skills taxonomies. The O*NET database maintained by the United States Department of Labor describes approximately nine hundred occupations linked to more than nineteen thousand task statements across some two thousand detailed work activities [49]. The European ESCO classification describes 3,039

occupations and 13,939 skills across twenty-eight languages, with an official correspondence to O*NET [50]. Both express occupations as compositions of skills and activities, which supplies an external basis for claims about occupational adjacency and a source for authoring occupationally representative tasks.

Evaluation benchmarks. For automated assessment, MT-Bench and the associated arena methodology [44], JudgeBench [47] and RewardBench [51] provide difficulty-calibrated reference sets and, more importantly, an established reporting methodology. For generated instructional content, EQGBench [52] and EduBench [53] provide protocols for evaluating generation quality. Text-based interactive environments including TextWorld [54] and Jericho [55] serve as established settings for evaluating agents in branching narrative.

Measurement practice. Two constructs require separate instruments and should not be reported as a single accuracy figure. Generation quality concerns whether produced content preserves required domain facts and elicits the intended constructs, and is assessed by expert review against specified criteria. Assessment accuracy concerns agreement between automated and expert judgment of the same response, and is properly reported as Cohen’s kappa, stratified by item difficulty, alongside test-retest consistency. The distinction matters because the three failure modes — biased judgment, unstable judgment, and judgment that fails selectively on difficult items — are separately diagnosable only when reported separately. As to expected magnitude, the closest methodological precedent, a multi-agent architecture performing unobtrusive assessment in a serious game, reports correlations between behavioural estimates and external outcomes in the range of 0.28 to 0.33, with near-zero correlation against prior attainment [56]. The latter figure is as informative as the former, since it evidences discriminant validity. Moderate correlations are the realistic expectation for open-ended behavioural assessment.

2.7 Responsibility considerations

Three considerations arise from the intended user population and the nature of the data involved, each corresponding to a requirement of the project’s specification.

Data protection for young users. A system addressed partly to secondary and tertiary students processes behavioural and performance data concerning individuals who may be minors, engaging Vietnam’s personal data protection regime and comparable frameworks for users elsewhere. Aggregated statistics derived from individual performance carry a residual privacy surface, since aggregate presentations can under some conditions permit inference about individuals.

Fairness in automated assessment. An automated judgment of a user’s demonstrated competence may influence that user’s estimation of their own capability. The evaluator biases documented in Section 2.5 are therefore not solely accuracy concerns but fairness concerns, and the corresponding controls — chance-corrected validation, cross-family evaluation, rubric anchoring — function as fairness safeguards. Presenting assessment formatively, and not as a determination affecting access to opportunity, keeps the consequences proportionate to the reliability the method can currently support.

Accuracy of occupational representation. The realistic job preview evidence carries an ethical as well as a pedagogical implication: a system’s value depends on representing occupations candidly, including their disadvantages, and is undermined if representation is shaped by commercial interest. Expert validation of

authored material, discussed in Section 2.5, is the principal safeguard against a system conveying an inaccurate picture of a real profession with apparent authority.

2.8 Summary

The background supports a coherent position. Performed, honest representation of occupational work conveys information that description does not, and this effect generalises across domains (2.1). Competence can be inferred from behaviour provided the assessment is constructed so that target constructs are necessary to task completion, and such competence is meaningfully transferable across related occupations (2.2). Game-based delivery sustains engagement, but its contribution to competence is marginal, so skill development must rest on task authenticity rather than reward structures, and randomisation must serve variety rather than reward scheduling (2.3). Career theory supports treating occupational fit as changing and as situated within a whole life, while explaining why self-report alone is a weak basis for guidance (2.4). Generative and assessment technologies make scenario production and open-response evaluation feasible, subject to explicit grounding, validated judgment, and expert oversight (2.5), and established taxonomies and benchmarks permit both grounding and evaluation without bespoke data collection (2.6). The user population imposes data protection, fairness and accuracy obligations that are design constraints rather than subsequent considerations (2.7).

Two qualifications in this evidence bear directly on what any such system may claim. The first is the marginal effect of gamification on competence [24]; the appropriate response is to locate skill development in authentic task performance and validated assessment rather than in reward mechanisms. The second is Sitzmann's finding that simulation exceeds comparison instruction only as a supplement, and underperforms as a sole method [27]; a system of this kind is therefore properly positioned as informing occupational choice and building initial self-efficacy, not as a substitute for employment or formal instruction. Both qualifications are treated in what follows as constraints on claims rather than as objections to be set aside.

3 Related Work

Chapter 2 established what the evidence indicates should be effective. This chapter examines what has been built. Section 3.1 reviews research prototypes; Section 3.2 reviews deployed systems, internationally and in Vietnam; Section 3.3 derives comparison criteria from the attributes the background identifies as consequential for learning systems, applies them, and states the resulting gap. Throughout, attention is given both to the limitations of existing work and to the design solutions worth adopting from it.

3.1 Related work from research literature

A growing body of research applies games and generative models to career development. It establishes the direction as active and credible, and three limitations recur across it.

Reflection as the terminal outcome. The most developed prototypes assist users in thinking about a career rather than performing one. CareerSim [57] is a language-model-driven career development game combining generated content with alumni data and career construction theory, advancing an avatar through life stages to produce a personalised report; a preliminary study with twelve participants reported improved understanding and reflection. Its interaction concerns a career trajectory rather than the tasks constituting an occupation, and the authors note limited contextual depth and difficulty with distant scenarios. Future You [58], in which participants conversed with a model-generated future self, reduced anxiety and increased future self-continuity across 344 participants, and Letters from Future Self [59] found comparable benefits for career exploration and resilience with thirty-six participants. Both demonstrate measurable psychological benefit from structured imagination of a future self. Neither provides the basis on which a user might discover, through performance, that the imagined self would find the work unsuitable — the informational deficit identified in Section 2.1. The design element worth adopting is the prospective framing itself, which these studies show is motivationally effective and which can be applied at the entry point of an experiential system rather than as its substance.

Trajectory abstraction without task fidelity. A second group models career development under uncertainty while abstracting from the work. CareerPooler [60] represents career development through a physical metaphor in which generated events perturb a trajectory, and reports significant gains in engagement, information gain, satisfaction and career clarity relative to a conversational baseline across twenty-four participants. It constitutes the strongest available evidence that generated, interactive career exploration outperforms conversational advice, and its treatment of unplanned events accords with the theoretical position in Section 2.4. Because the representation is metaphorical, however, the user does not perform occupational tasks, and outcomes are consequently reported in terms of clarity rather than demonstrated capability. Its event-generation pipeline, which balances favourable and unfavourable outcomes to avoid a uniformly positive trajectory, is directly adaptable to any system incorporating stochastic career events. Comparable abstraction characterises generative career imagination for children [61] and career simulation for self-regulation in secondary students [62].

Guidance without measurement. A third group improves the guidance interaction while leaving assessment unaddressed. A career counselling agent grounded in self-determination theory [63] reduced decision-making difficulty and improved engagement, confirming the applicability of the motivational framework in Section 2.3; its outcomes are measured through self-report instruments rather than through anything the user demonstrably did. Among the systems reviewed here, none infers competence from behaviour within a task, and none validates automated judgment against an expert-constructed standard. The techniques required are established outside the career domain — memory-consistent interactive narrative [40][43], rubric-based automated evaluation with its attendant validation requirements [44][46][47], and unobtrusive behavioural assessment [56] — but have not been combined within it.

Evaluation practice in this literature is a further limitation. Sample sizes range from twelve to thirty-six, with one exception at 344, and designs are predominantly qualitative or within-subject. Such designs are appropriate for establishing feasibility and eliciting user response, but they do not support claims about assessment accuracy, which requires comparison against an external standard.

3.2 Related work from deployed systems

3.2.1 International systems

Authored task simulation. Forage [64] is the established platform in experiential career exposure: employer-authored simulations in which users attempt representative tasks and compare their work against a model answer, distributed without charge and funded through employer talent acquisition. It substantiates the underlying premise that users will voluntarily attempt occupational tasks, and its task decomposition — discrete, self-contained work products with supporting context — is a directly adaptable model for scenario structure. Its constraints follow from its authoring model. Content expands only as employers commission it; repetition yields identical material, limiting exploration to a single pass; and assessment terminates in a fixed model answer, which demonstrates what a competent response looks like without evaluating the response the user produced. The catalogue also reflects its funding, concentrating in fields that sponsor recruitment. Human-mediated alternatives such as Virtual Internships and Extern provide genuine workplace exposure at correspondingly limited scale, while career video and mentoring platforms provide exposure without performance.

Generated task simulation and workplace assessment. A more recent group generates content rather than commissioning it. EntryLevel [65] produces AI-generated job simulations parameterised by field, skill and difficulty with an accompanying automated tutor, establishing commercially that occupational task content can be generated; it delivers these as discrete projects without progression or a connecting structure between occupations. Anthropos [66] conducts extended workplace scenarios in which a user interacts with several model-driven colleagues and produces work artefacts, explicitly generating behavioural evidence rather than self-report ratings. It is the closest deployed analogue to behavioural assessment as described in Section 2.2, and its multi-role scenario construction is adaptable. Its orientation is nonetheless towards personnel selection: the evidence generated serves an employer’s hiring decision rather than a user’s exploration, which determines both what is measured and who receives it. A third group — including conversation and interview practice systems with adaptive personas and scored feedback [67] — rehearses the selection process rather than the work, and its scoring dimensions, being conversational, do not generalise to occupational task performance.

Generative narrative systems. The systems that have most successfully solved the generation problems described in Section 2.5 are not career systems. AI Dungeon [68] and character.ai [69] established substantial demand for model-driven interactive narrative, particularly among the demographic most relevant to career exploration, and equally demonstrate the consistency degradation that unconstrained generation produces at scale. Hidden Door [70] is the more instructive case: it generates narrative within bounded authored components rather than through open generation, with the reported consequence that its narratives remain coherent across sessions. This is independent commercial corroboration of the finding in Section 2.5 that authored structure, not model capability, governs consistency, and its architecture is the most directly adaptable element identified in this review. What these systems lack is any representational commitment to a real domain, any model of competence, and any assessment.

Assessment and orientation instruments. O*NET and its associated public interfaces, together with commercial and institutional counterparts, constitute the most

widely used category. They administer an inventory, return ranked occupational suggestions, and terminate. Their occupational databases are a genuine asset — and, as noted in Section 2.6, a usable one — but the guidance model rests on the self-report foundation whose limitations Section 2.4 describes.

3.2.2 Vietnamese systems

The domestic market is materially homogeneous, which is itself the relevant finding.

Assessment-based provision. JobWay [71], reported in Vietnamese government press as the first digitised career orientation application in the country, comprises interest and personality assessment, descriptions of approximately three hundred occupations drawn from international sources, and linked institutional information. Notably, its occupational descriptions include the common difficulties of each occupation — an implicit application of the realistic job preview principle, delivered as text. JobTest.vn [72] is the most instrumentally developed, offering multiple validated inventories alongside an aptitude instrument adapted and standardised for Vietnamese use and an occupational database of substantial scale; its sophistication is concentrated in the measurement of self-report. Hướng nghiệp Sông An [73] combines counsellor training with a library of licensed instruments, representing the most methodologically careful domestic provision and remaining entirely assessment- and counsellor-based. Hướng Nghiệp LwL [74] advertises experiential activity alongside assessment and occupational information for a younger audience, though its substance remains assessment, content and mentoring.

Generative systems in the domestic market. Recent entrants apply generative models to career guidance, uniformly in conversational form. AI Hay [75] provides a model-driven agent predicting admission likelihood and recommending institutions and fields, together with an automated career-personality instrument; a parallel development across Vietnamese universities has produced admissions advisory agents at institutional scale. These establish market acceptance of AI-assisted career guidance domestically. None provides performed experience of occupational work, and none assesses demonstrated capability. Among the systems surveyed, no Vietnamese system offers generated, performed occupational simulation — a claim that should be re-verified close to submission, given the rate of change in this market.

3.3 Comparison, gap analysis, and positioning

3.3.1 Derivation of comparison criteria

Comparing systems against the features of a proposed system produces a circular result. The criteria below are instead derived from attributes that Chapter 2 identifies as determining whether a system of this general class — one intended to help a person learn about and choose an occupation — achieves its purpose. Each is stated as a general property, applicable to any system in the category, and each traces to an evidential basis rather than to a design preference.

The first concerns the nature of what the user does. Experiential learning and simulation transfer research (2.1) indicate that engagement with representative tasks of actual practice conveys understanding that description does not; task authenticity is therefore a primary attribute. The second concerns what is conveyed. The

realistic job preview literature (2.1) and the decision-difficulty taxonomy (1.1) establish that balanced information, including unattractive aspects, is what improves fit — hence completeness and candour of occupational representation. The third concerns responsiveness: adaptive instruction and the generative literature (2.5) indicate that content which responds to the individual outperforms fixed content, giving adaptivity to the learner. The fourth concerns whether the system knows what the user can do: evidence-centered design (2.2) establishes both the possibility of inferring competence from behaviour and the conditions under which such inference is valid, giving validity of competence measurement. The fifth follows from it — assessment is educationally inert without actionable response, giving quality of feedback and guidance. The sixth derives from the gamification and self-determination literatures (2.3): sustained engagement determines whether any learning occurs at all. The seventh concerns navigation: career theory and competency-based progression (2.2, 2.4) indicate that occupational choice involves comparing and moving among related options, giving support for pathway navigation. The eighth concerns durability: the authoring literature (2.5) shows that content about real occupations degrades unless it can be extended and maintained, giving scalability and currency of content. A ninth attribute, the existence of empirical validation, is treated in the discussion rather than tabulated, since it concerns the evidence for a system rather than a property of it.

3.3.2 Comparative assessment

Ratings use three levels, defined as follows. Full (●) indicates the attribute is a core designed property of the system. Partial (◐) indicates it is present but limited in scope, incidental, or a by-product of another objective. Absent (○) indicates it is not provided. Ratings reflect the systems as surveyed and would require revision as products change; per advisor guidance, the criteria set and ratings are to be finalised once the project scope is fixed.

Table 3.1: Comparison of system classes against the derived criteria.

Attribute	Authored simulation (Forage)	Generated simulation (En-tryLevel, thropos)	sim- (En- An- sim.ai etc.)	Interview practice (career-sim.etc.)	Generative nar- games (Hidden Door etc.)	Assessment & orientation (O*NET, VN market)	Research proto- types (Career-Sim, Career-Pooler)	Proposed system
C1 Task authenticity	• employer-authored real work products	• generated tasks	• role	⊖ interview talk, not work	⊖ fictional worlds	⊖ inventory only	⊖ trajectory, not tasks	• tasks drawn from occupational taxonomies
C2 Candour of occupational representation	• includes unglamorous tasks	⊖ varies by generation	⊖ not its purpose	⊖ no real occupations	⊖ text lists difficulties	⊖ events skew narrative	• candour is a design requirement	
C3 Adaptivity to the learner	⊖ fixed content, identical on repeat	• parameterised generation	• adaptive personas	• fully generative	⊖ branching by score	• generated per user	• seeded generation per user	
C4 Validity of competence measurement	⊖ model answer, no judgment	⊖ behavioural evidence, unvalidated	⊖ scored, conversational only	⊖ none	⊖ self-report only	⊖ self-report outcomes	⊖ designed; validity to be established	
C5 Quality of feedback and guidance	⊖ exemplar, not response-specific	⊖ automated tutor	• per-dimension coaching	⊖ none	⊖ generic occupation reports	⊖ reflective prompts	• result, feedback and guidance separated	
C6 Sustained engagement	⊖ one-pass certificate	⊖ discrete projects	⊖ practice utility	• high replayability	⊖ single sitting	⊖ short study exposure	⊖ designed; retention untested	
C7 Support for pathway navigation	⊖ isolated simulations	⊖ unconnected projects	⊖ single-role focus	⊖ world, not career, traversal	• ranked occupational lists	⊖ linear life stages	• competence-linked adjacency	
C8 Scalability and currency of content	⊖ commissioned per employer	• generated on demand	• generated on demand	• generated on demand	⊖ periodic database updates	⊖ prototype scale	• generation with expert validation	

• Full — a core designed property ⊖ Partial — limited, incidental, or a by-product ⊖ Absent. Annotations state the basis for each rating. Ratings for the proposed system denote design intent; those marked ⊖ are targeted but not yet empirically established.

3.3.3 Gap and positioning

The assessment indicates no systematic deficiency in any single system so much as a consistent failure to combine attributes that have been demonstrated separately. Authored simulation platforms achieve task authenticity and candour but cannot adapt or scale, and do not assess. Generated simulation systems achieve authenticity, adaptivity and scale, and where they assess behaviour they do so for selection rather than exploration, without pathway structure. Generative narrative systems achieve adaptivity and sustained engagement at high quality while making no representational commitment to real occupations. Assessment and orientation instruments — including the entire Vietnamese market surveyed — achieve pathway navigation and scale while omitting performance altogether. Research prototypes achieve adaptivity and engagement but stop short of task performance and competence measurement, and their evaluations do not support accuracy claims.

The consequent gap is not a missing feature but an unrealised combination: performed, candid occupational tasks that adapt to the individual, from which competence is measured with validated automated judgment, connected by a structure that relates occupations through the competencies they share, and maintained through expert authoring. Each constituent has been demonstrated — task authenticity by Forage, generation and behavioural evidence by EntryLevel and Anthropos, coherent generative narrative by Hidden Door, pathway structure by occupational taxonomies, engagement by the research prototypes — and their integration has not.

The present project addresses this combination. Its design adopts specific solutions identified above: bounded authored components as the means of maintaining generative consistency, discrete self-contained task structure for scenario construction, multi-role scenario composition for eliciting interpersonal competencies, balanced event generation for representing career uncertainty, and prospective framing at the entry point. Its contribution beyond assembly lies in the assessment and evaluation layer, where the requirements established in Section 2.5 — chance-corrected agreement, cross-family evaluation, difficulty stratification — are applied to a class of system in which such validation has not previously been reported. The claims the project may make are correspondingly bounded by Section 2.8: engagement and clarity effects are supported by existing evidence, whereas competence gain is a hypothesis to be tested against the reference set rather than a property the design confers.

4 The Proposed System

4.1 Business Context

4.1.1 Application domain

The application domain is career guidance and exploration, situated at the intersection of educational technology and occupational information provision. Participants in this domain are individuals facing an occupational decision, institutions supporting them, and the bodies that publish occupational information. The domain's characteristic activity is the transfer of occupational information to a person

who must act on it, and Chapter 1 established that this transfer is currently incomplete: guidance provision is dominated by instruments that elicit self-description rather than convey what work involves.

The proposed system, JobQuest, addresses this by allowing a user to perform representative tasks of an occupation, receive an assessment of the competencies those tasks elicited, and use that assessment to determine which further occupations to explore. It is a platform for informed exploration rather than a placement, recruitment, or credentialing service, and this boundary determines several of the business rules below.

4.1.2 Business processes

Three processes constitute the operation of the platform. They are modelled formally in Section 5.1.1.

The exploration process is the primary user-facing process. A registered user optionally completes an orientation conversation, which is provided for users who do not know which occupation to begin with and is not a prerequisite. The user selects an occupation and a level within it, is presented with the role, team, and working context before any task begins, and then works through the tasks comprising that level. Each task is a scenario drawn from the occupation and is answered either in a structured form or in free text. The system evaluates each response, records the competencies it evidences, and returns a result, feedback specific to the response, and guidance for improvement. Completing a level awards competence credit; accumulated credit unlocks higher levels of the same occupation or, where competencies overlap sufficiently, entry to an adjacent occupation. On completing an occupation, the user may rate it on five attributes, which contribute to that occupation’s aggregate profile.

The content lifecycle process governs how occupations enter the catalogue. A domain expert authors a seed for an occupation, specifying the required facts of the scenario, the competencies it must elicit, and the rubric against which free responses are judged. The system generates candidate scenarios from the seed. The expert reviews these, and a content administrator publishes the occupation once review is complete. Published content is subject to periodic revalidation, since occupational practice changes.

The evaluation process is internal and supports the research objectives. Expert-authored reference items — scenarios paired with gold-standard judgments — are assembled into a reference set, against which generation quality and assessment accuracy are measured offline as specified in Section 2.6.

4.1.3 Business rules

Table 4.1: Business rules governing the proposed system.

ID	Rule	Origin
BR-01	The entry level of every occupation is accessible without prerequisite.	Exploration is the purpose; gating discovery defeats it

ID	Rule	Origin
BR-02	Access to a higher level requires the competence credit specified for that level.	Competency-based progression (Section 2.2)
BR-03	Entry to an adjacent occupation requires demonstrated competencies overlapping that occupation's entry requirements, determined from the occupational taxonomy.	Section 2.2, 2.6
BR-04	An occupation may be rated only by a user who has completed it.	Rating validity depends on exposure
BR-05	No scenario seed or rubric is published without domain-expert validation.	Section 2.5; accuracy of occupational representation
BR-06	Assessment results are formative. They are not transmitted to employers and do not determine access to any external opportunity.	Section 2.7; proportionality to assessment reliability
BR-07	Occupational representation must include the unfavourable aspects of the work.	Realistic job preview evidence (Section 2.1)
BR-08	Statistics derived from user performance are published only in aggregate form.	Section 2.7
BR-09	The automated assessor must be drawn from a different model family than the scenario generator.	Self-preference bias (Section 2.5)

4.1.4 Organisational policies and legal-ethical considerations

The platform processes personal data, including performance data, of users who may be minors, and is therefore subject to Vietnam's personal data protection regime and to comparable frameworks for users in other jurisdictions. Three policies follow. Data minimisation limits collection to what the exploration and assessment functions require. Aggregate-only publication (BR-08) governs any statistic derived from user performance, with suppression thresholds applied so that small populations cannot be de-anonymised. Formative-use limitation (BR-06) prevents

assessment output from acquiring consequences the method’s reliability does not support.

Two further policies concern the system’s use of generative models. Provenance disclosure requires that users be informed that scenarios are machine-generated from expert-authored material and that assessments are machine-produced. Expert accountability assigns responsibility for the occupational accuracy of published content to the validating domain expert rather than to the generating model. Detailed processing policies, including third-party model providers, are specified in Section 5.3.10.

4.2 System Description

4.2.1 System overview

JobQuest is a web-based platform that presents occupations as sequences of performable tasks, assesses the competencies a user demonstrates while performing them, and uses those assessments to structure further exploration. Its purpose is to supply occupational information experientially, at a breadth that authored content cannot reach, to people who must choose or change an occupation.

The experience is delivered through three presentation and progression conventions selected against the evidence in Section 2.3 rather than by preference. Progression follows a role-playing convention, in which levels and access to further occupations are earned through demonstrated competence rather than time spent, which the role-play evidence supports as the genre most associated with skill acquisition. Scenarios are presented in a branching narrative form comparable to a visual novel, the narrative moderator being among the strongest identified in the gamification meta-analyses. Career and life events are introduced probabilistically across sessions, a convention drawn from roguelike design, so that the circumstances bearing on occupational fit vary between runs; consistent with the harm evidence in Section 2.3, this randomisation governs narrative variety only and is not coupled to any reward schedule or monetisation. These conventions carry engagement; consistent with Section 2.8 they are not relied upon for competence development, which rests on task authenticity and assessment.

The system serves four user groups.

The explorer is the primary user: a student choosing a field of study, a graduate entering the labour market, or a worker considering transition. The explorer performs tasks, receives assessment and guidance, accumulates competence credit, and navigates among occupations. All other groups exist to make this possible.

The domain expert is a practitioner or educator with substantive knowledge of a specific occupation. The expert authors scenario seeds and rubrics, reviews generated scenarios for occupational accuracy, and revalidates content as practice changes. This group is the guarantor of the system’s central claim, namely that its representations correspond to real work.

The content administrator manages the occupation catalogue as a body of content: commissioning and scheduling expert authoring, tracking review status, controlling publication, and monitoring coverage and currency across occupations. In the terms of the report template this is the business administrator, and its context-specific designation here is content lead.

The system administrator operates the platform: configuring and monitoring the generation and assessment models, managing capacity and cost, administering accounts and access, and overseeing data protection controls.

4.2.2 Research problems

The project addresses five research problems, derived from the objectives in Section 1.2 and the evidence in Chapter 2.

RP-1, constrained generation. Given an expert-authored seed specifying required facts and target competencies, how should generation be constrained so that produced scenarios remain occupationally accurate and continue to elicit the target competencies? The literature establishes that fluent generation is readily obtained and that fidelity requires explicit constraint, but does not specify the constraint for this content type.

RP-2, validity of automated assessment. To what extent does automated rubric-based judgment of free-text responses agree with domain-expert judgment, measured with chance correction and stratified by item difficulty? Section 2.5 establishes that reported agreement in this literature is systematically optimistic and that difficulty-stratified performance is generally unreported.

RP-3, competence-based adjacency. How should demonstrated competencies be mapped onto occupational adjacency so that an unlock decision is defensible rather than arbitrary? Occupational taxonomies supply skill compositions but not thresholds for readiness.

RP-4, engagement without displacement. Which engagement mechanisms sustain continued exploration without displacing the task authenticity on which learning depends? Section 2.3 establishes that gamification affects competence only marginally, so the mechanisms must be selected against that constraint.

RP-5, evaluation methodology. How can a reference set be constructed, at feasible cost within a single occupational field, that supports substantive claims about RP-1 and RP-2 without recourse to large-scale user study?

4.2.3 Challenges and difficulties

Generation fidelity is the principal technical difficulty. Occupational content is factual, and a scenario that is plausible but inaccurate is worse than no scenario, since it conveys a confident false impression of a real profession. Long-horizon consistency compounds this within multi-task levels.

Assessment validity is the principal methodological difficulty. Section 2.5 indicates that agreement figures for automated judgment are frequently overstated and that performance degrades unevenly with difficulty; obtaining defensible figures requires a reference set that is stratified and expert-authored, which is expensive to produce.

Expert authoring cost constrains breadth. Each occupation requires expert time for seed authoring, rubric specification, and review. This is the reason for restricting the first phase to a single field, and it makes the authoring tooling a first-class component rather than an administrative convenience.

Cold start affects the aggregate occupational profiles, which require a population of completions before they are informative. Until that population exists, profiles

are absent or unreliable, and the interface must represent this honestly rather than display an unfounded figure.

Evaluation without deployment limits what the project can establish. Offline benchmarking substantiates generation quality and assessment accuracy but cannot establish engagement retention or effects on career decisions, which require longitudinal study. The claims are bounded accordingly, consistent with Section 2.8.

Localisation presents a further difficulty: occupational taxonomies are authored in English for other labour markets, and their occupational structures do not map cleanly onto Vietnamese practice, requiring expert adaptation rather than translation.

4.3 User's Requirements

Requirements are expressed first as user stories per group, then analysed into functional, non-functional, and data requirements. Each requirement carries a priority (Must, Should, Could), an assessment of feasibility within the project's scope and schedule, and a statement of how it is to be verified.

4.3.1 Functional requirements

User stories — explorer:

- As an explorer, I want to try the entry level of any occupation without prerequisite, so that I can survey possibilities before committing attention to one.
- As an explorer, I want to be told my role, team, and working context before a task begins, so that my decisions are situated rather than abstract.
- As an explorer, I want to perform tasks that practitioners would recognise, so that what I learn reflects the occupation.
- As an explorer, I want to see the unfavourable aspects of an occupation, so that my impression is not a recruitment pitch.
- As an explorer, I want my response assessed with a result, an explanation, and guidance, so that I understand both the judgment and how to improve.
- As an explorer, I want competencies I have demonstrated to be recorded, so that my progress reflects what I can do rather than time spent.
- As an explorer, I want to move to an adjacent occupation when my competencies qualify me, so that I need not restart when exploring related work.
- As an explorer, I want to rate an occupation after completing it and see how others rated it, so that I can compare my impression with a broader one.
- As an explorer, I want a private record of what I have attempted and demonstrated, so that my exploration accumulates.

User stories — domain expert:

- As a domain expert, I want to author a seed specifying required facts, target competencies, and a rubric, so that generated scenarios remain accurate.
- As a domain expert, I want to review generated scenarios before publication and reject or correct them, so that inaccurate material does not reach users.
- As a domain expert, I want to see where the automated assessor disagrees with my judgment, so that I can refine the rubric.

User stories — content administrator:

- As a content administrator, I want to add a new occupation and track its progress through authoring, review, and publication, so that catalogue growth is managed.
- As a content administrator, I want to see coverage and last-validated dates across occupations, so that I can identify stale content.

User stories — system administrator:

- As a system administrator, I want to configure which models perform generation and assessment, so that the separation required by BR-09 is enforced and providers can be changed.
- As a system administrator, I want to monitor usage, cost, and failures, so that the service remains operable within budget.

Table 4.2: Functional requirements with priority, feasibility, and verification method.

ID	Functional requirement	Group	Priority	Feasibility	Verification
FR-01	Register and authenticate a user account	Explorer	Must	High	Account creation and sign-in succeed; unauthenticated access to progress is refused
FR-02	Present the entry level of any published occupation without prerequisite	Explorer	Must	High	Entry level reachable for a new account with zero credit
FR-03	Present role, team, and working context before the first task of a level	Explorer	Must	High	Context screen precedes first task in every level

ID	Functional require- ment	Group	Priority	Feasibility	Verification
FR-04	Generate a task scenario from a published seed, preserving its required facts	Explorer	Must	Medium	Expert review of sampled generations against the seed’s fact list
FR-05	Accept structured responses and score them against an answer key	Explorer	Must	High	Known-input tests produce expected scores
FR-06	Accept free-text responses and score them against the seed’s rubric	Explorer	Must	Medium	Agreement with expert judgment on the reference set (Cohen’s kappa)
FR-07	Return result, response-specific feedback, and forward guidance as distinct elements	Explorer	Must	High	All three present and distinguishable for every assessed response
FR-08	Record competencies evidenced by each assessed response	Explorer	Must	Medium	Competence records correspond to rubric criteria for sampled sessions

ID	Functional require- ment	Group	Priority	Feasibility	Verification
FR-09	Award com- petence credit on level completion and unlock levels per BR-02	Explorer	Must	High	Threshold boundary tests unlock at and only at the specified credit
FR-10	Offer entry to adjacent occupa- tions when overlap conditions are met per BR-03	Explorer	Should	Medium	Offered set matches taxonomy- derived expectation for seeded profiles
FR-11	Conduct an optional orientation conversa- tion producing candidate occupa- tions	Explorer	Should	Medium	Conversation completes and yields a non-empty ranked candidate set
FR-12	Collect a five- attribute rating on completion and display aggregates	Explorer	Should	High	Rating accepted only after completion; aggregate updates; suppres- sion below threshold
FR-13	Present a private record of attempted occupa- tions, competen- cies, and awards	Explorer	Should	High	Record reflects session history for a seeded account

ID	Functional require- ment	Group	Priority	Feasibility	Verification
FR-14	Introduce probabilistic career and life events affecting subsequent exploration	Explorer	Could	Medium	Events occur within configured probability bounds across repeated runs
FR-15	Author and edit a seed with required facts, target competencies, and rubric	Expert	Must	High	Seed persists and is retrievable with all three components
FR-16	Review, approve, reject, or correct generated scenarios	Expert	Must	High	Unapproved scenarios are not served to explorers
FR-17	Report disagreements between automated and expert judgment	Expert	Should	Medium	Disagreement list matches recomputation over the reference set
FR-18	Manage occupation lifecycle: create, track status, publish, retire	Administrator	Must	High	State transitions follow the defined lifecycle; publication requires completed review
FR-19	Report catalogue coverage and last-validated dates	Administrator	Should	High	Report matches stored metadata

ID	Functional require- ment	Group	Priority	Feasibility	Verification
FR-20	Configure generation and assessment model providers, enforcing separation	Administrator	Must	Medium	Configuration rejects identical provider for both roles
FR-21	Monitor usage, cost, and error rates	Administrator	Should	High	Metrics present and consistent with generated load

4.3.2 Non-functional requirements

Table 4.3: Non-functional requirements with priority, feasibility, and verification method.

ID	Non- functional require- ment	Category	Priority	Feasibility	Verification
NFR-01	Scenario generation completes within 5 s at the 95th percentile; a progress indicator is shown beyond 2 s	Performance	Must	Medium	Latency distribution measured over at least 200 sampled generations
NFR-02	Assessment of a free-text response completes within 8 s at the 95th percentile	Performance	Must	Medium	Latency distribution measured over at least 200 sampled assessments

ID	Non-functional requirement	Category	Priority	Feasibility	Verification
NFR-03	Session progress survives interruption and resumes without loss	Reliability	Must	High	Forced interruption tests resume at the recorded position
NFR-04	On model provider failure, no committed progress is lost and the user is notified within 10 s	Reliability	Should	Medium	Fault injection against the provider interface; zero progress-record loss across injected failures
NFR-05	Personal data is collected only as required, stored encrypted, and deletable on request	Security and privacy	Must	High	Data inventory review; deletion request removes records
NFR-06	Aggregate statistics apply a minimum-population suppression threshold	Security and privacy	Must	High	Aggregates below threshold are withheld
NFR-07	Machine generation and machine assessment are disclosed to the user	Transparency	Must	High	Disclosure present at the point of use

ID	Non-functional requirement	Category	Priority	Feasibility	Verification
NFR-08	Interface language and occupational content support Vietnamese and English	Usability	Should	Medium	Both locales render complete content
NFR-09	At least 80% of representative readers correctly state a task's objective after reading its scenario	Usability	Should	Medium	Comprehension check with a minimum of 20 representative readers
NFR-10	Adding an occupation requires no code modification	Maintainability	Must	High	A new occupation is published through the authoring interface alone
NFR-11	Mean model cost per completed session remains within the configured ceiling, with per-session cost recorded	Cost	Should	Medium	Cost telemetry aggregated per session and compared against the configured ceiling

NFR-12	Model provider is replaceable without changes to seeds or rubrics	Portability	Should	Medium	Provider substitution test with unchanged content
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4.3.3 Data requirements

Table 4.4: Data requirements with source, priority, feasibility, and verification method.

ID	Data requirement	Source	Priority	Feasibility	Verification
DR-01	Occupational reference data: occupations, constituent tasks, work activities, and skills	O*NET and ESCO, adapted for Vietnamese practice by domain experts	Must	Medium	Imported records resolve against source identifiers
DR-02	Competency taxonomy with definitions and observable indicators, distinguishing soft and technical competencies	Derived from DR-01 under expert adaptation	Must	Medium	Every rubric criterion resolves to a taxonomy entry
DR-03	Occupational adjacency relations with overlap measures	Computed from DR-01/DR-02	Should	Medium	Computed adjacency reviewed by experts for plausibility
DR-04	Scenario seeds: required facts, target competencies, narrative skeleton, rubric	Domain experts	Must	High	Seed schema validation; expert sign-off recorded

ID	Data requirement	Source	Priority	Feasibility	Verification
DR-05	Generated scenario instances with provenance linking to seed and model version	System	Must	High	Every served scenario carries resolvable provenance
DR-06	User profile and progress: competence credit, unlocked levels, occupation history	System	Must	High	Progress reconstructs from stored records
DR-07	Assessment records: response, score, competencies evidenced, feedback, guidance	System	Must	High	Records complete for every assessed response
DR-08	Occupation ratings on five attributes, with completion precondition enforced	Explorers	Should	High	Ratings exist only for completed occupations
DR-09	Expert reference set: scenarios with gold-standard judgments, stratified by difficulty	Domain experts	Must	Medium	Set covers defined strata with recorded inter-expert agreement

ID	Data re- quirement	Source	Priority	Feasibility	Verification
DR-10	Operational telemetry: latency, cost, error rates, model versions	System	Should	High	Telemetry present and queryable

Personal data within DR-06, DR-07, and DR-08 is subject to the policies in Section 4.1.4. DR-09 contains no personal data and is retained as a research asset.

4.4 Summary

This chapter specified the proposed system against the domain established in Chapters 1 to 3. The business context identified three operating processes — exploration, content lifecycle, and evaluation — and nine business rules, of which several derive directly from the evidence reviewed in Chapter 2 rather than from product preference: accessible entry levels, competence-gated progression, taxonomy-derived adjacency, mandatory expert validation, formative-only assessment, and separation of the generating and assessing models. The system description identified four user groups and stated five research problems and six substantive difficulties, the most consequential being generation fidelity, assessment validity, and expert authoring cost. The requirements analysis produced twenty-one functional, twelve non-functional, and ten data requirements, each prioritised, assessed for feasibility, and paired with a verification method. Chapter 5 models the processes and interactions these requirements imply.

5 Analysis and Design

5.1 Analysis

5.1.1 Business process modelling

The three processes identified in Section 4.1.2 are modelled below. Diagram sources are given in a text notation for reproduction in the project’s modelling tool.

Exploration process. The process begins with authentication and ends either in occupation completion and rating or in the user selecting a different occupation. The decision points are the optional orientation, the response type within a task, and the unlock evaluation on level completion.

flowchart TD

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A[Register / sign in] --> B{Knows which<br/>occupation to try?}
B -- No --> C[Orientation conversation]
C --> D[Candidate occupations presented]
B -- Yes --> E[Browse catalogue]
D --> E
E --> F[Select occupation and level]
F --> G[Present role, team, working context]
G --> H[Present task scenario]
H --> I{Response type}
I -- Structured --> J[Score against answer key]
I -- Free text --> K[Score against rubric]
J --> L[Record competencies evidenced]
K --> L
L --> M[Return result, feedback, guidance]
M --> N{More tasks<br/>in level?}
N -- Yes --> H
N -- No --> O[Award competence credit]
O --> P{Credit meets<br/>threshold?}
P -- Higher level --> Q[Unlock next level]
P -- Adjacent occupation --> R[Offer adjacent occupation]
P -- Neither --> S[Suggest further practice]
Q --> T{Occupation<br/>complete?}
R --> T
S --> T
T -- Yes --> U[Collect five-attribute rating]
U --> V[Update aggregate profile]
T -- No --> E
V --> E

```

Figure 5.1: Exploration process.

Content lifecycle process. This process governs occupational accuracy and is the operational expression of BR-05.

flowchart TD

```

A[Administrator commissions occupation] --> B[Expert authors seed:<br/>facts, competencies, r
B --> C[System generates candidate scenarios]
C --> D[Expert reviews generations]
D --> E{Accurate and<br/>eliciting?}
E -- No --> F[Expert corrects seed or rejects scenario]
F --> C
E -- Yes --> G[Expert approves]
G --> H[Administrator publishes occupation]
H --> I[Available to explorers]
I --> J{Revalidation<br/>due?}
J -- Yes --> D

```

Figure 5.2: Content lifecycle process.

Evaluation process. This process is internal to the project and supports RP-1 and RP-2. Experts construct reference items stratified by difficulty; generation output and assessor output are compared against them; agreement is computed with chance correction and reported per stratum, together with test-retest consistency.

5.1.2 Use case modelling

Four actors correspond to the user groups in Section 4.2.1. Two external systems participate: the model provider, which performs generation and assessment, and the occupational data source, which supplies reference data.

flowchart LR

```
EX([Explorer]); DE([Domain expert]); CA([Content administrator]); SA([System administrator])
MP([Model provider]); OD([Occupational data source])
```

```
EX --- U1(UC-01 Explore occupation catalogue)
EX --- U2(UC-02 Complete orientation conversation)
EX --- U3(UC-03 Perform a task scenario)
EX --- U4(UC-04 Progress to a further level or occupation)
EX --- U5(UC-05 Rate a completed occupation)
EX --- U6(UC-06 Review personal record)
```

```
DE --- U7(UC-07 Author scenario seed and rubric)
DE --- U8(UC-08 Validate generated scenario)
DE --- U9(UC-09 Review assessment disagreements)
```

```
CA --- U10(UC-10 Manage occupation lifecycle)
CA --- U11(UC-11 Monitor coverage and currency)
```

```
SA --- U12(UC-12 Configure generation and assessment models)
SA --- U13(UC-13 Monitor operation and cost)
```

```
U3 --- MP
U8 --- MP
U4 --- OD
U7 --- OD
```

Figure 5.3: Use case diagram for the proposed system.

Tabular descriptions of the key use cases follow. UC-03 and UC-08 are the use cases on which the project’s research problems depend.

UC-03 — Perform a task scenario

Table 5.1: Use case UC-03, Perform a task scenario.

Field	Content
Actor	Explorer (primary); model provider (supporting)
Goal	Attempt a task representative of the occupation and receive an assessment of it
Precondition	The user is authenticated and the level is accessible under BR-01 or BR-02
Trigger	The user selects a level and proceeds past the context screen

Field	Content
Main flow	1. System retrieves the published seed for the task. 2. System generates a scenario instance preserving the seed's required facts. 3. System presents the scenario and the response affordance. 4. User submits a response. 5. System scores the response against the answer key or the rubric according to response type. 6. System records the competencies evidenced. 7. System returns result, response-specific feedback, and forward guidance. 8. System advances to the next task or concludes the level.
Alternative flow	4a. User abandons the task: progress to that point is recorded and the session may be resumed (NFR-03).
Exception flow	2a. Generation fails or violates seed constraints: system retries, then serves a previously validated scenario instance rather than an unvalidated one. 5a. Assessment fails: response is stored and queued; user is informed that assessment is pending.
Postcondition	An assessment record exists (DR-07) and competence records are updated (DR-06)
Business rules	BR-07 representation includes unfavourable aspects; BR-09 assessor differs from generator
Requirements	FR-03, FR-04, FR-05, FR-06, FR-07, FR-08; NFR-01, NFR-02, NFR-07

UC-04 — Progress to a further level or occupation

Table 5.2: Use case UC-04, Progress to a further level or occupation.

Field	Content
Actor	Explorer (primary); occupational data source (supporting)
Goal	Convert demonstrated competence into access to further exploration
Precondition	A level has been completed and competence credit awarded
Trigger	Level completion

Field	Content
Main flow	1. System computes accumulated competence against the next level's threshold. 2. If met, the next level is unlocked. 3. System computes overlap between demonstrated competencies and the entry requirements of adjacent occupations. 4. Occupations whose overlap conditions are satisfied are offered. 5. User selects a destination or returns to the catalogue.
Alternative flow	2a. Threshold not met: system indicates the competencies outstanding and suggests tasks that elicit them.
Exception flow	3a. Adjacency data unavailable: within-occupation progression is offered and the cross-occupation offer is suppressed rather than guessed.
Postcondition	Accessible levels and offered occupations reflect demonstrated competence
Business rules	BR-02, BR-03
Requirements	FR-09, FR-10; DR-02, DR-03

UC-08 — Validate generated scenario

Table 5.3: Use case UC-08, Validate generated scenario.

Field	Content
Actor	Domain expert (primary); content administrator (secondary)
Goal	Ensure published scenarios are occupationally accurate and elicit their target competencies
Precondition	A seed exists and candidate scenarios have been generated
Trigger	Generation completes, or revalidation falls due
Main flow	1. System presents a candidate scenario alongside its seed's required facts and target competencies. 2. Expert judges factual accuracy against the seed. 3. Expert judges whether the scenario necessarily elicits the target competencies. 4. Expert approves. 5. Scenario becomes eligible for publication.

Field	Content
Alternative flow	4a. Expert rejects the scenario: it is withheld and regeneration is requested. 4b. Expert corrects the seed: prior generations from that seed are invalidated and regenerated.
Exception flow	3a. Expert judges the scenario passable without the target competency: the seed is returned for revision, since the evidence model would otherwise be invalid (Section 2.2).
Postcondition	Only expert-approved scenarios are servable
Business rules	BR-05, BR-07
Requirements	FR-15, FR-16; DR-04, DR-05

UC-07 — Author scenario seed and rubric

Table 5.4: Use case UC-07, Author scenario seed and rubric.

Field	Content
Actor	Domain expert (primary)
Goal	Specify the material from which accurate scenarios can be generated and responses judged
Precondition	An occupation has been commissioned and reference data are available
Trigger	Assignment by the content administrator
Main flow	1. Expert selects the occupation and consults its tasks and skills from reference data. 2. Expert specifies required facts the scenario must preserve. 3. Expert specifies target competencies, each resolving to the competency taxonomy. 4. Expert specifies the narrative skeleton and the response affordances. 5. Expert specifies the rubric, each criterion attributed to a target competency. 6. System validates the seed against its schema and rejects criteria not attributable to a competency.
Exception flow	6a. Unattributable rubric criteria: seed is rejected with the offending criteria identified.
Postcondition	A schema-valid seed exists and is eligible for generation
Business rules	BR-05
Requirements	FR-15; DR-01, DR-02, DR-04

Field	Content
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UC-05 — Rate a completed occupation

Table 5.5: Use case UC-05, Rate a completed occupation.

Field	Content
Actor	Explorer
Goal	Record an impression of the occupation and contribute to its aggregate profile
Precondition	The user has completed the occupation (BR-04)
Trigger	Occupation completion
Main flow	1. System presents the five rating attributes. 2. User submits ratings. 3. System records the rating and recomputes the aggregate profile. 4. System presents the aggregate alongside the user's own ratings.
Alternative flow	2a. User declines to rate: completion is recorded without a rating.
Exception flow	3a. Population below the suppression threshold: aggregate is withheld and the condition is stated rather than displaying an unfounded figure.
Postcondition	Rating recorded; aggregate updated or suppressed
Business rules	BR-04, BR-08
Requirements	FR-12; NFR-06; DR-08

The remaining use cases — catalogue exploration, orientation conversation, personal record review, disagreement review, lifecycle management, coverage monitoring, model configuration, and operational monitoring — are specified in the same form in Appendix 1.

The traceability matrix below traces every requirement to the use case realising it, so that no requirement is left unrealised and no use case introduces behaviour the requirements do not specify.

Table 5.6: Traceability from requirements to use cases.

Use case	Functional	Non-functional	Data
UC-01 Explore catalogue	FR-02	NFR-08, NFR-09	DR-01
UC-02 Orientation conversation	FR-11	NFR-01, NFR-07	DR-01, DR-06
UC-03 Perform task scenario	FR-03, FR-04, FR-05, FR-06, FR-07, FR-08	NFR-01, NFR-02, NFR-03, NFR-04, NFR-07	DR-04, DR-05, DR-07

Use case	Functional	Non-functional	Data
UC-04 Progress further	FR-09, FR-10, FR-14	—	DR-02, DR-03, DR-06
UC-05 Rate occupation	FR-12	NFR-06	DR-08
UC-06 Review personal record	FR-13	NFR-05	DR-06, DR-07
UC-07 Author seed and rubric	FR-15	NFR-10	DR-01, DR-02, DR-04
UC-08 Validate scenario	FR-16	NFR-10	DR-04, DR-05
UC-09 Review disagreements	FR-17	—	DR-07, DR-09
UC-10 Manage lifecycle	FR-18	NFR-10	DR-04
UC-11 Monitor coverage	FR-19	—	DR-04
UC-12 Configure models	FR-20	NFR-11, NFR-12	DR-05
UC-13 Monitor operation	FR-21	NFR-11	DR-10
Account management	FR-01	NFR-05	DR-06

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